

of these principal-agent problems and has allowed the mutual fund industry to flourish.

Mutual funds have also encouraged people to save. People do indeed need to save (see chapter 5), and they would have saved far less without mutual funds. Individuals have gained access to investment skills that would be out of reach to them in running their own portfolios, and their funds are protected from the worst abuses by managers. Unfortunately, the benefit of investment skill accrues to the managers of the funds, on average, rather than to the investors in the funds. Mutual funds underperform, or at best equal, the market, and investors pile into funds with high past returns that aren't repeated going forward.

4. Closed-End Funds

Closed-end funds are ideal vehicles to hold illiquid assets because the number of shares is fixed and they are traded in the secondary market. That means there can be no redemption from the fund. So closed-end funds can harvest an illiquidity premium.⁴⁹ There is, however, a difference between the exchange price of closed-end shares and the underlying value of the fund's investments (the NAV). This difference can be large, is usually negative, and varies over time. This phenomenon is called the *closed-end fund discount puzzle*.⁵⁰

Figure 16.9 plots the closed-end fund discount from 1934 to 2012, which is an updated series from Baker, Wurgler, and Yu (2012). The last portion of this data is the closed-end fund discount index constructed by Thomas J. Herzfeld Advisors, a firm specializing in closed-end funds. The average discount is 11%, but there is considerable time variation.

Some reasons for this negative average discount are:

1. Taxes

The discount can arise from *tax overhang*.⁵¹ Managers of a mutual fund can realize capital gains at a time that is inopportune for investors. (This takes away investors' tax-timing option; see chapter 12.)

2. Irrationality

Lee, Shleifer, and Thaler (1991) argue that the closed-end fund discount is due to investor irrationality. The discount itself is used as an input into

⁴⁹ See Stein (2005). Note that the illiquidity premium across asset classes is small, but there are large differences between the returns of liquid and illiquid assets within asset classes; see chapter 11.

⁵⁰ See Cherkas (2012) for a summary article on closed-end funds. Some investment companies are economically similar to closed-end funds in the sense that they are listed and the number of shares is fixed. Berkshire Hathaway is a famous example. These are corporations and do not fall under the 40-Act.

⁵¹ Originally due to Malkiel (1977). Chay, Choi, and Pontiff (2006) estimate that each dollar of distributed capital gains shrinks the discount by 7 cents.